Fiscal Year:	FY 2026	**Task Last Updated:**	FY 05/15/2026
PI Name:	Lytle, Jason		
Project Title:	Spaceflight Thrombosis and Risk Factors (STAR)		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77058	**Congressional District:**	36
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2024 80JSC024NA001-FLAGSHIP2 2024 HERO Appendix E: NASA Human Research Program Flagship2 Opportunity
Start Date:	05/01/2025	**End Date:**	04/30/2031
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Co-Investigator Steven Grover, previously at the University of North Carolina - Chapel Hill, has moved to the Beth Israel Deaconess Medical Center in Boston (Ed., 5/13/26).		
COI Name (Institution):	Carter, Katrina (Wyle Laboratories, Inc.) Grover, Steven (Beth Israel Deaconess Medical Center - grant 80NSSC26K0600) Kramer, Larry (University of Texas Health Science Center at Houston - grant 80NSSC26K0598) Laurie, Steven (Wyle Laboratories, Inc.) Macias, Brandon (NASA Johnson Space Center) Young, Millennia (NASA Johnson Space Center) Moll, Stephan (University of North Carolina, Chapel Hill - grant 80NSSC26K0604)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Weightlessness causes stagnant blood flow through the left internal jugular vein (IJV) in some, but not all individuals, and may increase for venous thromboembolism (VTE). Because not all individuals presenting with stagnant IJV flow during weightlessness develop a thrombus, additional coagulability changes may be necessary for clot formation. The ability to identify, preflight and during early flight, those at greatest risk for developing flow abnormalities and pro-thrombotic changes during flight would enable individualized and targeted prophylactic treatment of only high-risk astronauts. Additionally, inflight use of anticoagulants is rare and the impact of these during weightlessness is not well defined. The purpose of this study is to identify anatomical factors and biomarkers, preflight and during early flight, that together would indicate increased risk for VTE. A single preflight magnetic resonance imaging (MRI) will be used to assess cerebral venous structures and anatomical variation that may cause venous impingement during weightlessness. Ultrasound assessments of IJVs will occur in various postures twice preflight and during spaceflight on flight day (FD) 7, 14, 30, 60, 75, and 150 and at the end of a 2-week anticoagulant treatment phase. Plasma and whole blood samples will be collected during each ultrasound session. A 2-week prophylactic treatment of anticoagulants will be prescribed at FD 60-75 and will stop thereafter. Standard laboratory assays will be used to assess targeted biomarkers. Target-agnostic plasma biomarker discovery will use a commercial plasma proteomic platform to measure levels of 5,400 unique protein biomarkers involved in various biologic systems contributing to thrombosis. Functional coagulation parameters will be assessed in plasma samples using calibrated automated thrombography. This project seeks to identify individuals most at risk of VTE in order to inform targeted countermeasure application.
Rationale for HRP Directed Research:	This project is directed because it contains highly constrained research.
Research Impact/Earth Benefits:	This research has direct relevance to healthcare on Earth, where venous thrombus (VT) development remains a major cause of morbidity and mortality across a wide range of clinical populations. By studying how weightlessness alters jugular venous flow, vascular anatomy, and blood coagulability, this project helps reveal fundamental mechanisms that contribute to clot formation. The tools and approaches here, such as advanced imaging-based risk assessment and comprehensive plasma biomarker discovery have the potential to inform earlier, more precise identification of patients at risk for VT development on Earth. Through these innovations, astronaut health research not only safeguards crew members in space but contributes to improved prevention and diagnosis on Earth.
Task Progress:	The NASA Institutional Review Board (IRB) and international Human Research Multilateral Review Board have approved both an expedited pilot study and a full research study. Three crewmembers have signed informed consent for the pilot study and initiated data collection preflight and inflight. During preflight baseline data collection, crew participate in a brain MRI and a session in the NASA Cardiovascular and Vision Laboratory to assess ultrasound measures of internal jugular veins and thoracic duct. Venous blood samples are acquired for assessment of blood viscosity and frozen for future batch analysis of various biomarkers. The same ultrasound measures and venous blood sampling are planned for flight day (FD) 7, 14, 30, 75, 150, and R-30. There is no postflight testing.
Bibliography Type:	Description: (Last Updated:)